Autoclave Validation Format Approval

	WRITTEN BY:	REVIEWED BY:
Signature:		
Position:	SENIOR ANALYST	QC HEAD
Date:		

	APRROVED BY:	AUTHORIZED BY:
Signature:		
Position:	QA MANAGER	SUPT. PHARMACIST
Date:		

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EQUIPMENT DESCRIPTION:

1. **DELAMA AUTOCLAVE**- Industrial double door autoclave	2. **PHARMALAB AUTOCLAVE**: Industrial double door autoclave
Equipment Identification Number: BL/AC/PM/001	**Equipment Identification Number:** BL/AC/PM/002
Location: Production floor	**Location: Production floor**
Purpose To sterilize the filled pouches	

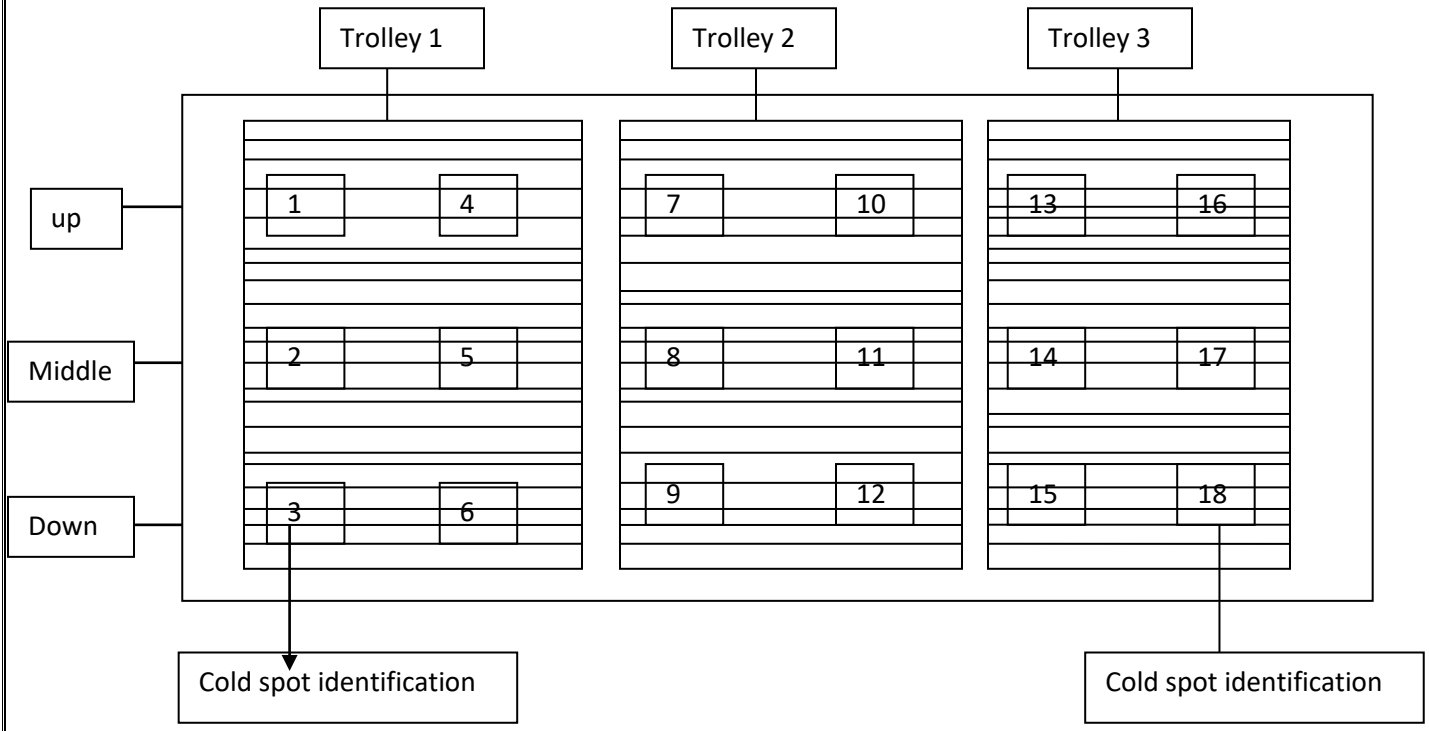
2.0 **VALIDATION TEAM:**

- Autoclave Operator.
- Production Supervisor
- Microbiologist.
- Chemist
- Technical Officer
- Quality Control Head
- Chief Engineering
- Quality Assurance Manager

3.0 **RESPONSIBILITIES:**

Autoclave Operator	Machine operation and execution of the protocol.
Production supervisor	Supervision of the operators and equipment
Supervisor Engineering Department.	To conduct heat distribution and penetration studies.
Microbiologist.	Preparation of the protocol To conduct the Biological indicator test/ chemical Strip. Documentation of Results
Quality Control Head	Execution of the protocol. To verify and document the results gathered during the validation of Sterilizer
Chief Engineer	Utility support
QA Manager	Approval
Supt. Pharm./ Factory Manager	Authorization

4.0 SCHEMATIC DIAGRAM OF DELAMA AUTOCLAVE WITH LOCATION OF TEMPERATURE SENSORS (INSIDE VIEW)



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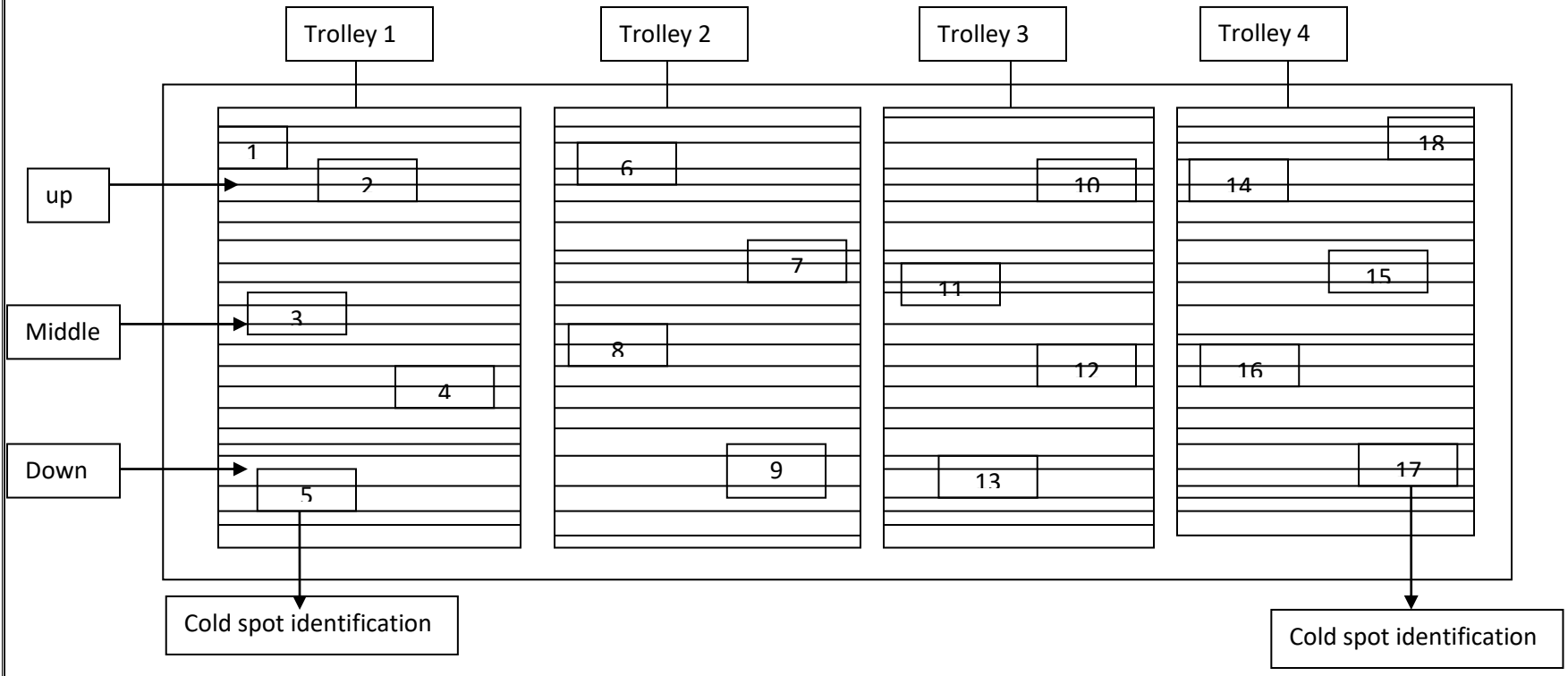
5.0 SCHEMATIC DIAGRAM OF PHARMALAB AUTOCLAVE WITH LOCATION OF TEMPERATURE SENSORS (INSIDE VIEW)



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6.0 VALIDATION OF AUTOCLAVE (CHECKLISTS REPORT) **DELAMA AUTOCLAVE** ☐ **PHARMALAB AUTOCLAVE** ☐

S. No.	Critical parameters to be met	Acceptance Criteria	Observation
1.	Verify all gauges & instruments are ok Temp. Chart / Recorder. Temp. Controller. Pressure Gauges.	All records must be within the acceptable limits	
2.	**Empty Chamber Heat Distribution Cycle. (No. of cycles: 3)** Set temperature: 116 - 118°C. Sterilization time: 40mins. Pressure bar: 1.5 – 1.7	Throughout the Sterilization phase all temperatures must be within 3°C range	
3.	**Machine parts:** - Safety valves for chamber, streamline ,air inlet - Pressure gauge for chamber, streamline, air inlet - Air pressure regulator - Non return valves - Heat Exchanger - Door gasket - Pressure switch/ pressure transmitter - Temperature sensor - Chart recorder - Programmable logic control (PLC)	Must conform with set parameters	

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	working		
4.	**Autoclaving with pouches** Set temperature: 116 - 118°C. Sterilization time: 40 mins. Biological indicators should be used in all 3 Cycles. Chemical indicators should be used in all 3 Cycles.	Throughout the sterilization phase all temperatures are within 2°C range. The colour of the Biological indicator with samples taken from the identified cold spot should not change in the inoculation broth after the incubation period of 7 to 14days at 45°C. Initial White colour of chemical indicator placed at the identified cold spot should change to black and cross from reject zone to accept zone of the strip	
5.	**Cold spot identification** During sterilization cycle, arranging the probes across the loaded chamber has the goal of assuring sterilization is attained across the load	Although the cold spot might be lagging before reaching the sterile hold temperature of 116°C. It must not drop below 116°C sterile hold temperature throughout the sterile hold time of 40mins.	

7.0 VALIDATION REPORT **PHARMALAB AUTOCLAVE (HEAT DISTRIBUTION STUDY)** ☐

DELAMA AUTOCLAVE (HEAT DISTRIBUTION STUDY) ☐

S/N	PARAMETERS	ACCEPTANCE CRITERIA	HEAT DISTRIBUTION STUDY (NO LOAD)			HEAT PENETRATION STUDY (FULL LOAD)		
			Trial 1	Trial 2	Trial 3	Trial 1	Trial 2	Trial 3
1.	Thermalog Chemical strips Point 1 Point 2 Point 3 Point 4 Point 5 Point 6 Point 7 Point 8 Point 9 Point 10 Point 11 Point 12 Point 13 Point 14 Point 15 Point 16 Point 17 Point 18	Chemical Strips must change colour and move from unsafe to safe zone						
2.	Microbiological Test Kits Point 1 Point 2 Point 3 Point 4							

	Point 5 Point 6 Point 7 Point 8 Point 9 Point 10 Point 11 Point 12 Point 13 Point 14 Point 15 Point 16 Point 17 Point 18	Test kit must be sterile after autoclave						
3.	Heating	Average Temperature difference between probe and thermometer must not be more than ± 2°C						
4.	Sterility test(sampled from cold spot)	Must be sterile						
5.	Bacteria Endotoxin	≤ 0.25IU/ml or less than 1.15^{0}c sum temperature rise for three rabbits.						

8.0 VALIDATION REPORT **PHARMALAB AUTOCLAVE (HEAT PENETRATION STUDY)** ☐
DELAMA AUTOCLAVE (HEAT PENETRATION STUDY) ☐

100ML ☐ **500ML** ☐ **1000ML.** ☐

S/N	PARAMETERS	ACCEPTANCE CRITERIA	HEAT PENETRATION STUDY (FULL LOAD)		
			Trial 1	Trial 2	Trial 3
1.	Thermalog Chemical strips Point 1 Point 2 Point 3 Point 4 Point 5 Point 6 Point 7 Point 8 Point 9 Point 10 Point 11 Point 12 Point 13 Point 14 Point 15 Point 16 Point 17 Point 18	Chemical Strips must change colour and move from unsafe to safe zone			
2.	Microbiological Test Kits Point 1 Point 2 Point 3 Point 4				

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	Point 5 Point 6 Point 7 Point 8 Point 9 Point 10 Point 11 Point 12 Point 13 Point 14 Point 15 Point 16 Point 17 Point 18	Test kit must be sterile after autoclave			
3.	Heating	Average Temperature difference between probe and thermometer must not be more than ± 2°C			
4.	Sterility test(sampled from cold spot)	Must be sterile			
5.	Bacteria Endotoxin	≤ 0.25IU/ml or less than 1.15^{0}c sum temperature rise for three rabbits.			

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9.0 VALIDATION REPORT PHARMALAB AUTOCLAVE (PERFORMANCE) ☐

DELAMA AUTOCLAVE (PERFORMANCE) ☐

FIRST FIRSTBATCH

DATE: _____

PRODUCT NAME: _____

BATCH NUMBER: _____

S/N	PARAMETERS	SPECIFICATION	RESULT
1.	Temperature		
2.	Hold time		
3.	Pressure		

SECOND BATCH

DATE: _____

PRODUCT NAME: _____

BATCH NUMBER: _____

S/N	PARAMETERS	SPECIFICATION	RESULT
1.	Temperature		
2.	Hold time		
3.	Pressure		

THIRD BATCH

DATE: _____

PRODUCT NAME: _____

BATCH NUMBER: _____

S/N	PARAMETERS	SPECIFICATION	RESULT
1.	Temperature		
2.	Hold time		
3.	Pressure		

_____	_____	_____
Technical Officer	Production Pharmacist	Quality Assurance Officer

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Remark:

_____

Certification:

_____ _____ _____

Engineering Manager Production Manager QA Manager

_____

Factory Manager/ Supt. Pharmacist

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